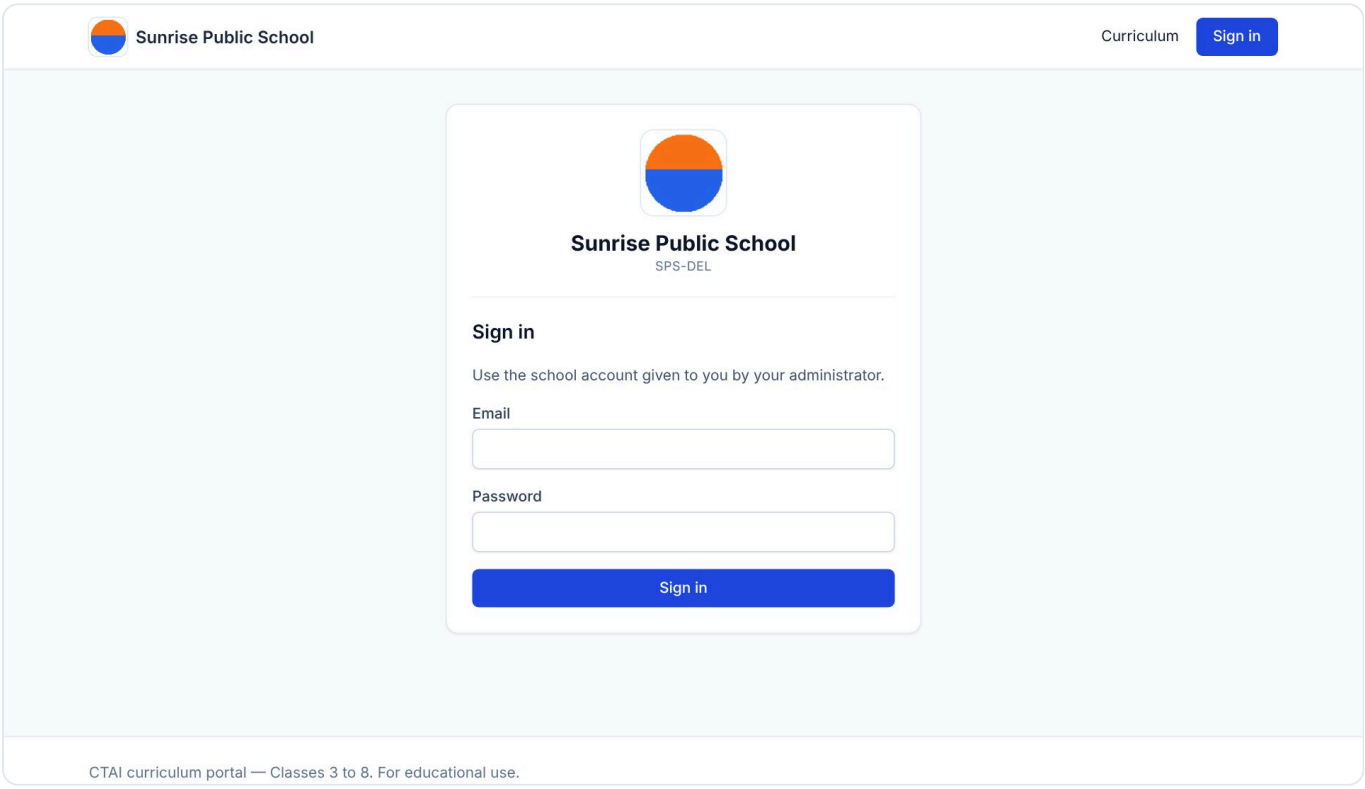
PRODUCT GUIDE · OCTOBER 2026

CTAI Portal Complete Guide

Coding, Computational Thinking & AI for Classes 3–8 — what the app does, how each person uses it, what is finished and what still needs to be done.



Every school gets its own branded sign-in page at **/<school-code>**.

Who this is for

School owners, principals, teachers and whoever runs the server.

App version

Branch `claude/cbse-ctai-curriculum-app-kyooB`, commit `057c849`

Curriculum basis

CBSE CT & AI Student and Teacher Handbooks, Classes 3–8, 2026-27 (first edition, March 2026)

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Reading this guide

Sections 1–3 explain the idea. Sections 4–9 walk through each person's screens with real screenshots of the app (section 8 is the classroom presentation mode). Sections 10–13 cover content, grading, safety and hosting. Sections 14–15 are the status report: what's finished, and a prioritised list of what is left.

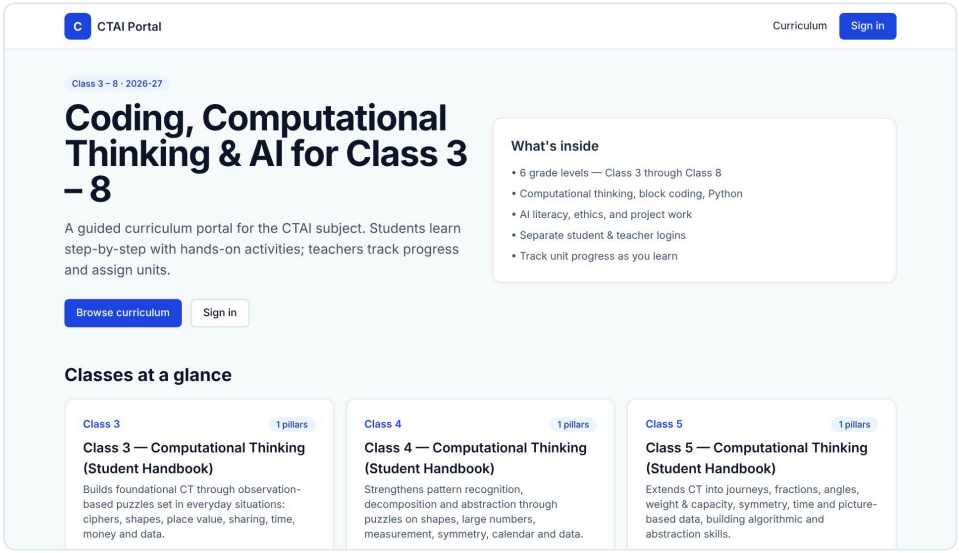
About the screenshots

All screenshots are from the current version running with made-up demo data — a school called “Sunrise Public School”, teacher “Ms. Roy”, students “Aarav”, “Diya”, “Kabir”, “Meera” and “Riya”. None of these are real people or accounts.

1 The portal at a glance

The CTAI Portal is a website where schools teach the new CBSE **Computational Thinking & AI** subject for Classes 3 to 8. Students read worked examples and answer practice questions that are marked instantly. Teachers decide when each chapter opens and see every student's progress. Many schools can share one installation, each with its own name, logo and sign-in page.

6 classes (3 to 8)	**80** chapters (68 CT + 12 AI)	**802** auto-graded practice questions	**160** worked examples
80 teacher-notes blocks (lesson plans)	**4** kinds of user	**∞** schools per installation	**2FA** for the portal owner



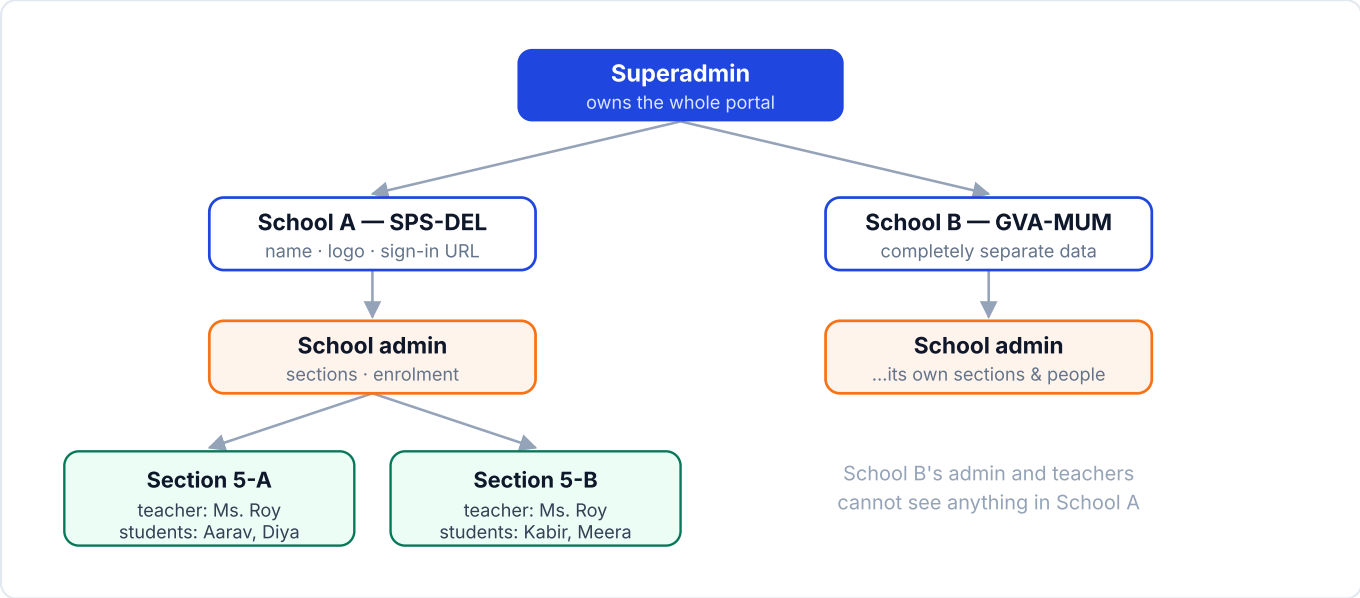
Home page. Anyone can browse the class list and chapter outlines; signing in unlocks practice and progress.

Built with

Part	Technology	Why it matters to you
Website & server	Next.js 14 (React) on Node.js	One app serves both the pages and the data; runs on any Node host.
Database	SQLite — a single file, `portal.db`	No database server to manage. Back up one file and you have everything.
Sign-in	Encrypted session cookie, hashed passwords, Google-Authenticator codes	Passwords are never stored in readable form.
Styling	Tailwind CSS	Clean layout that works on laptops, tablets and phones.

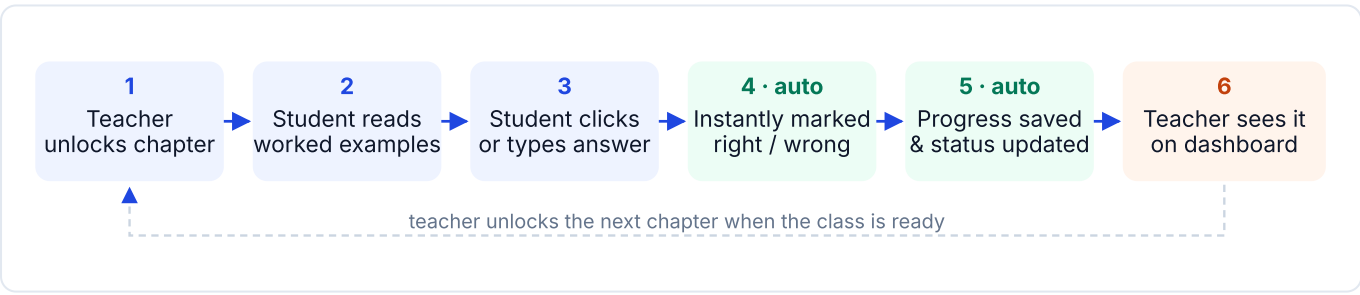
2 How everything fits together

Think of the portal as a tree. The portal owner adds schools; each school's admin sets up sections and enrols people; teachers run their own sections.



The learning loop

This is what happens every time a chapter is taught. Steps 4 and 5 happen automatically — no one has to mark anything.



3 Who can do what

There are four kinds of user. Each sees only what they need — and nothing from other schools.

Superadmin

- Adds, renames and removes schools
- Uploads school logos
- Creates each school's admin
- Turns on 2-step sign-in

Signs in at **/login**

School admin

- Creates sections (5-A, 5-B …)
- Assigns class teachers and co-teachers
- Enrols teachers and students
- Resets passwords

Signs in at **/<school-code>**

Teacher

- Sees only their own sections
- Unlocks chapters per section
- Tracks each student's progress
- Reads lesson plans (teacher notes)

Signs in at **/<school-code>**

Student

- Sees only their own class
- Opens chapters the teacher unlocked
- Practises with instant marking
- Watches their progress grow

Signs in at **/<school-code>**

Permission matrix

Action	Superadmin	School admin	Teacher	Student
Create / delete schools, upload logos	✓	—	—	—
Create school admins	✓	—	—	—
Create sections, assign teachers	—	✓ own school	—	—
Enrol teachers and students, reset their passwords	—	✓ own school	—	—
Unlock / lock chapters	—	no screen yet	✓ own sections	—
See student progress	—	counts only	✓ own sections	✓ self
Read teacher notes / lesson plans	✓	✓	✓	—
Answer practice questions (progress saved)	—	—	try only	✓ own class
Change own password	✓	✓	ask admin	ask admin
Two-step sign-in (Google Authenticator)	✓	—	—	—

Nobody can sign themselves up

Accounts are created top-down. The only exception is the very first visit to `/register`, which creates the superadmin once and then switches itself off for good.

4 Signing in

School users — `/<school-code>`

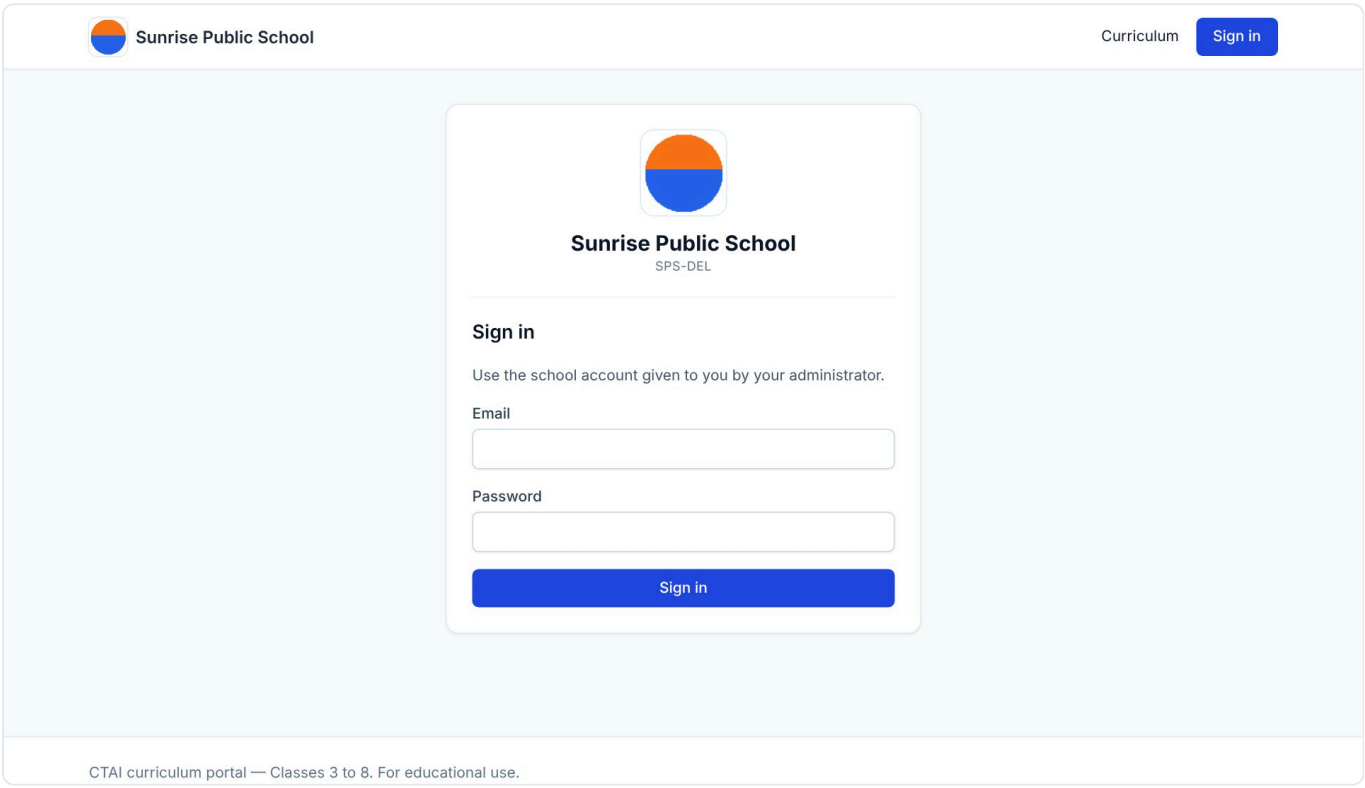
Each school has a branded page, for example `/SPS-DEL`. The school's logo and name appear in the header and above the sign-in box. Share this link with the school's staff and students.

An account from another school is refused here, so a student cannot accidentally sign in to the wrong school.

Portal owner — `/login`

The superadmin uses the plain `/login` page. If two-step sign-in is on, a second screen asks for the 6-digit code from Google Authenticator (or Authy).

After sign-in everyone lands on their own dashboard automatically.

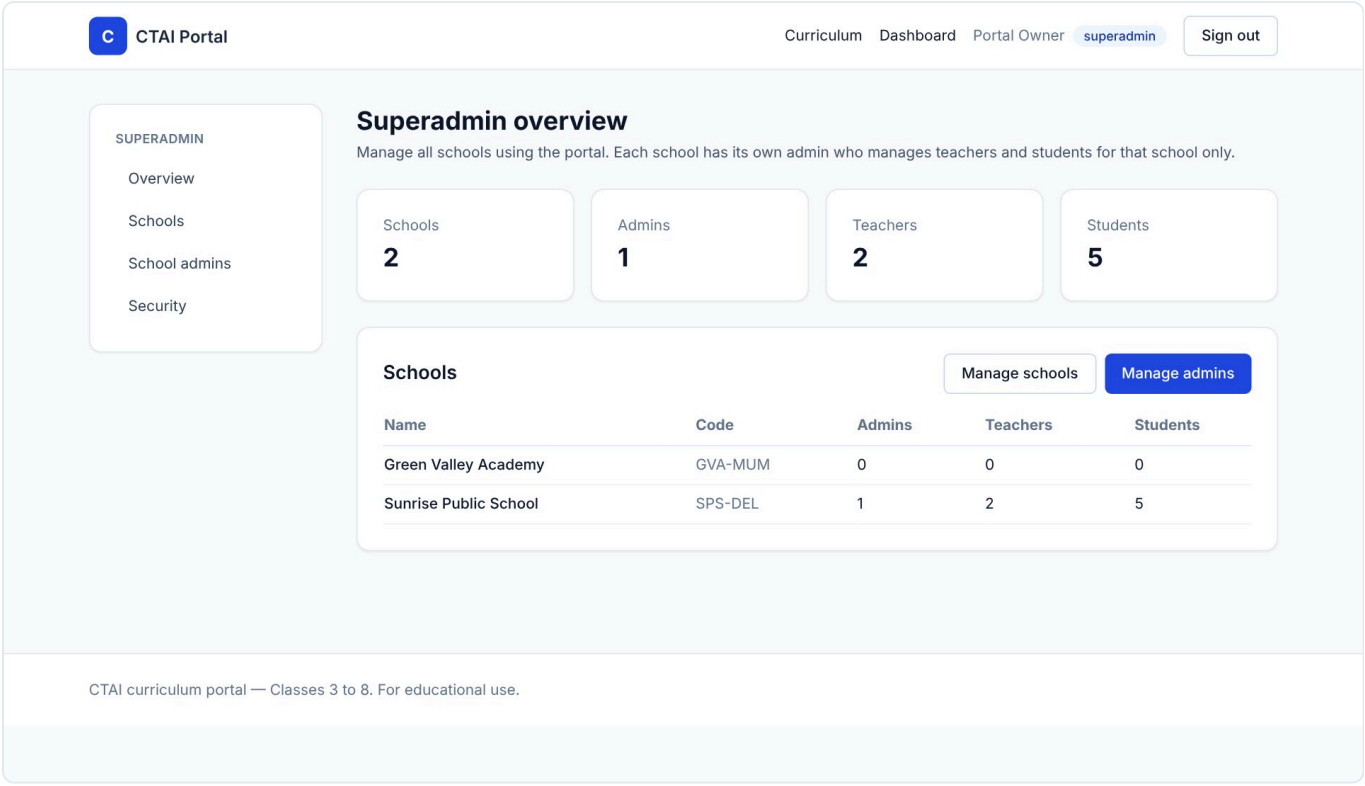


School sign-in page at `/SPS-DEL`, branded with the school's uploaded logo.

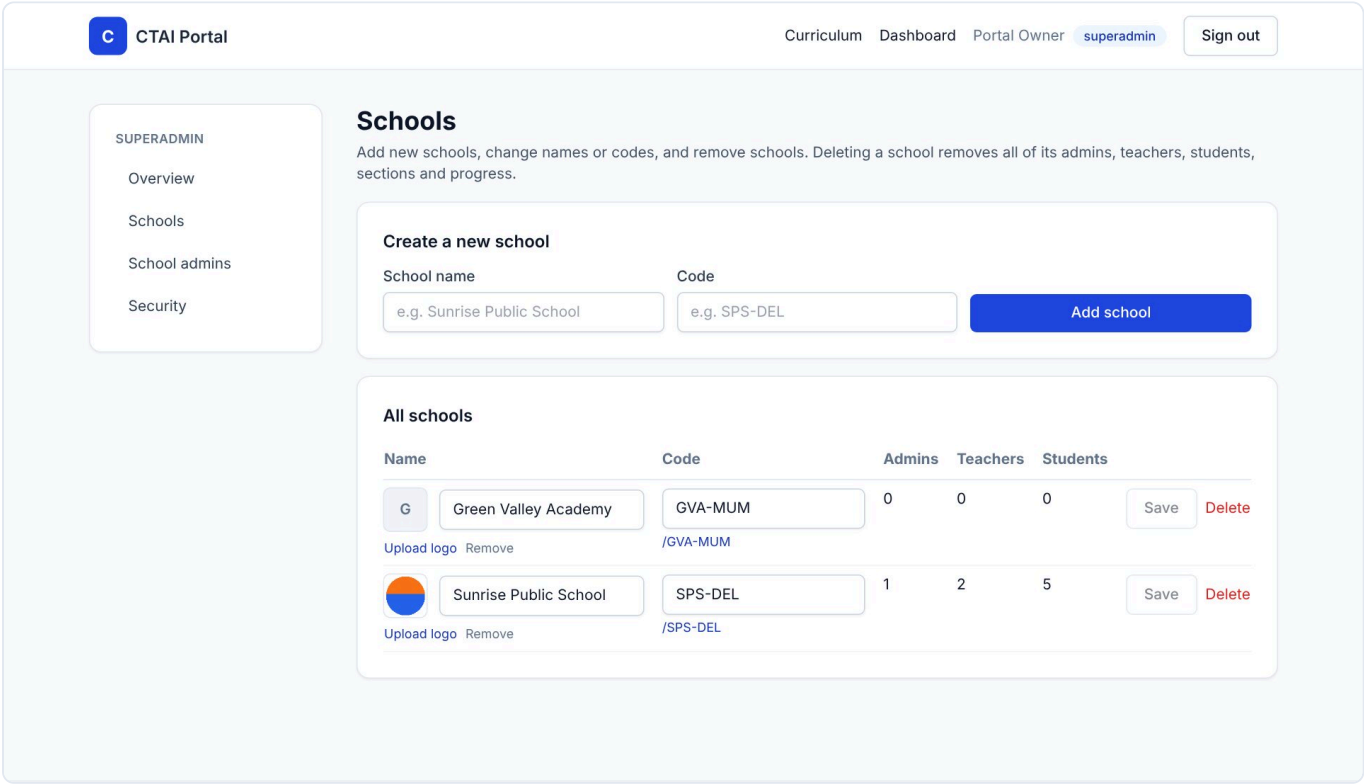
> **Forgotten passwords**
> There is no “forgot password” email yet. Students and teachers ask their school admin, who sets a new one from **Teachers & students → Reset**. School admins ask the superadmin. (See the to-do list, section 15.)

5 Superadmin (portal owner)

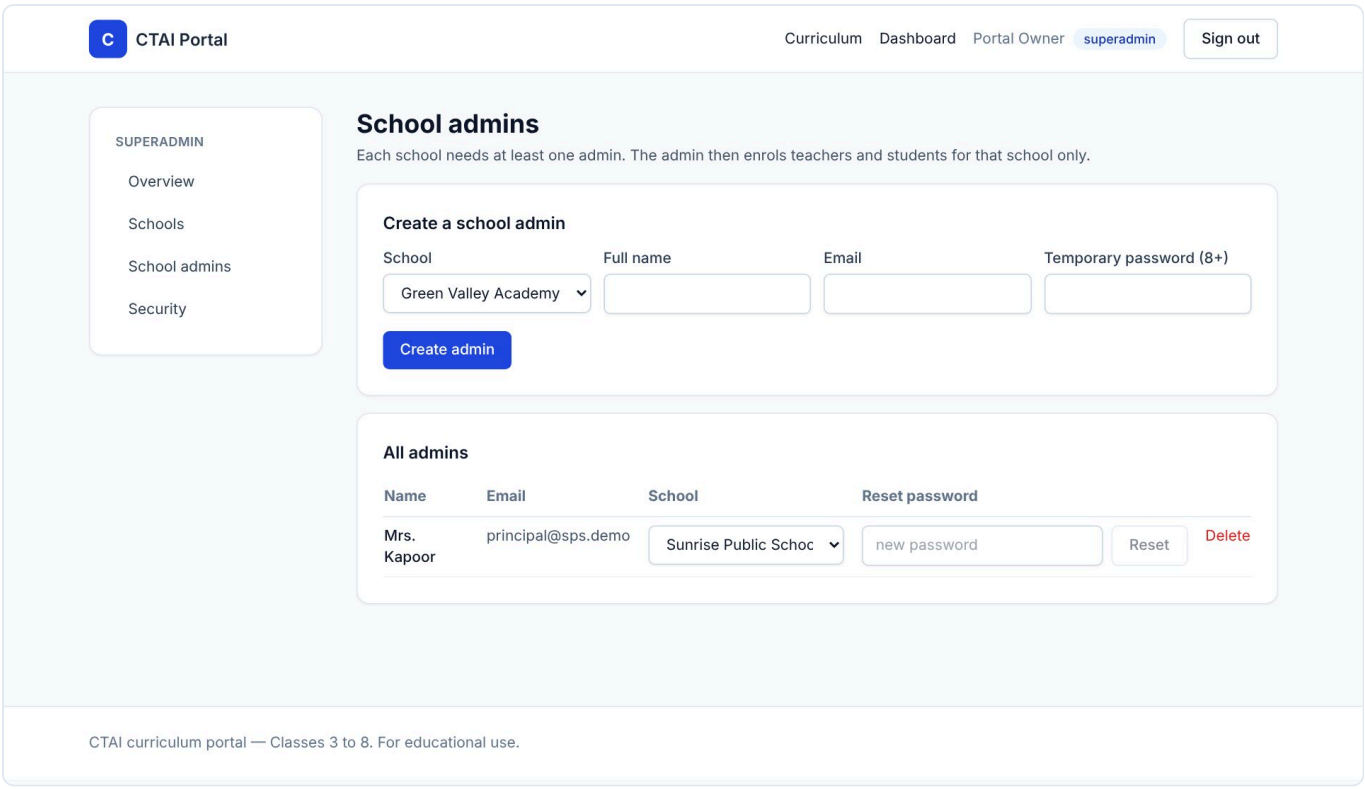
The superadmin runs the installation: adds schools, gives each one an admin, and keeps the owner account secure.



Overview — totals across every school, with a per-school breakdown of admins, teachers and students.

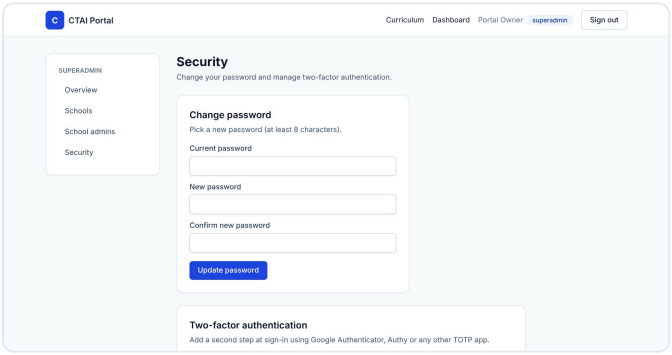


Schools — add a school with a name and short code. The code becomes the school's sign-in address (shown under each code). Upload a PNG/JPG/WebP/SVG logo up to 1 MB; schools without one show their initial.

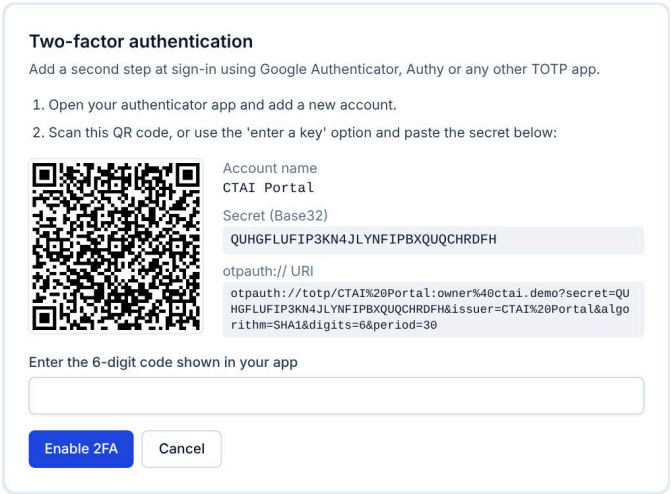


School admins — create the admin for each school, move them between schools, reset their password, or remove them.

Security page



Change password — needs the current password; new one must be 8+ characters.



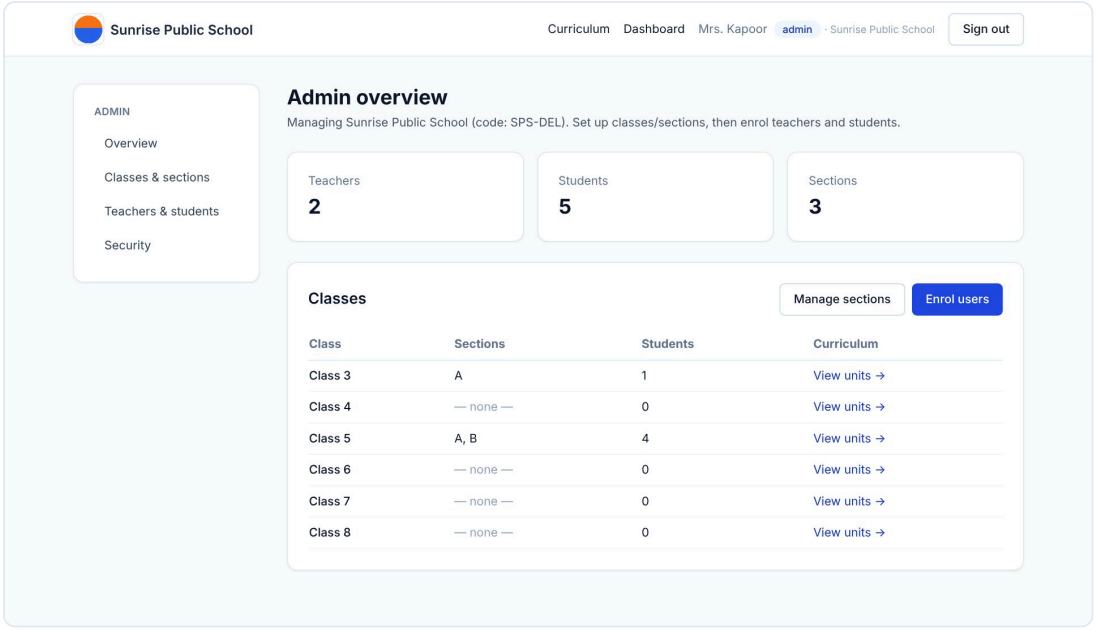
Two-step sign-in — scan the QR with Google Authenticator, then enter a code to switch it on. The QR is generated on your own server.

Deleting a school is permanent

It removes that school's admins, teachers, students, sections and all progress. There is no undo — keep backups (section 13).

6 School admin (principal)

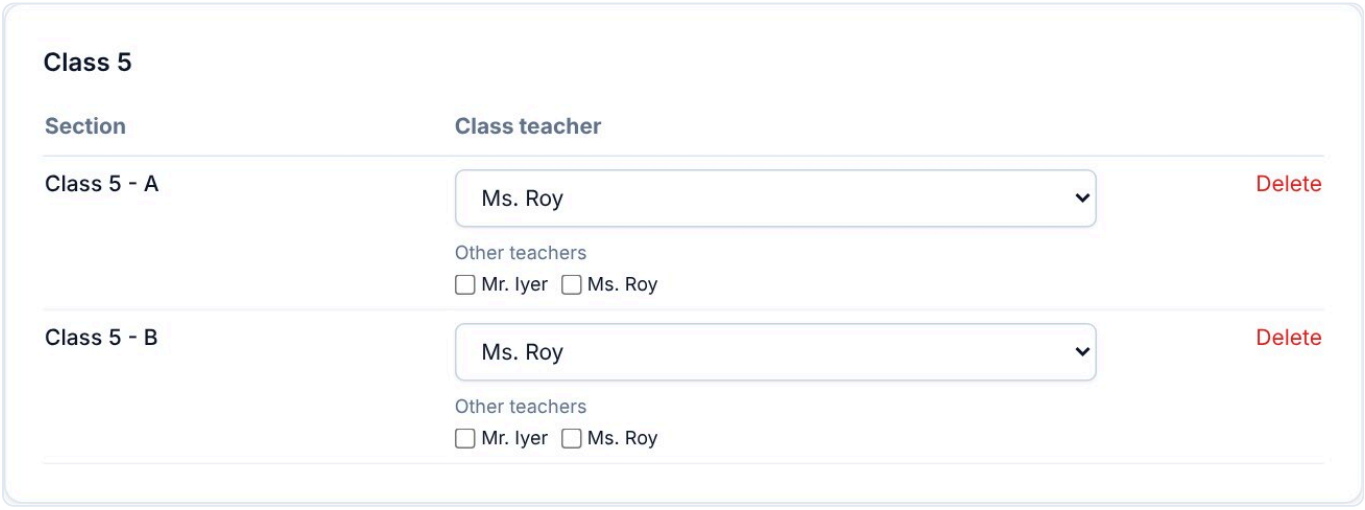
The school admin sets up the school once at the start of the year, then mainly enrols new students and resets passwords.



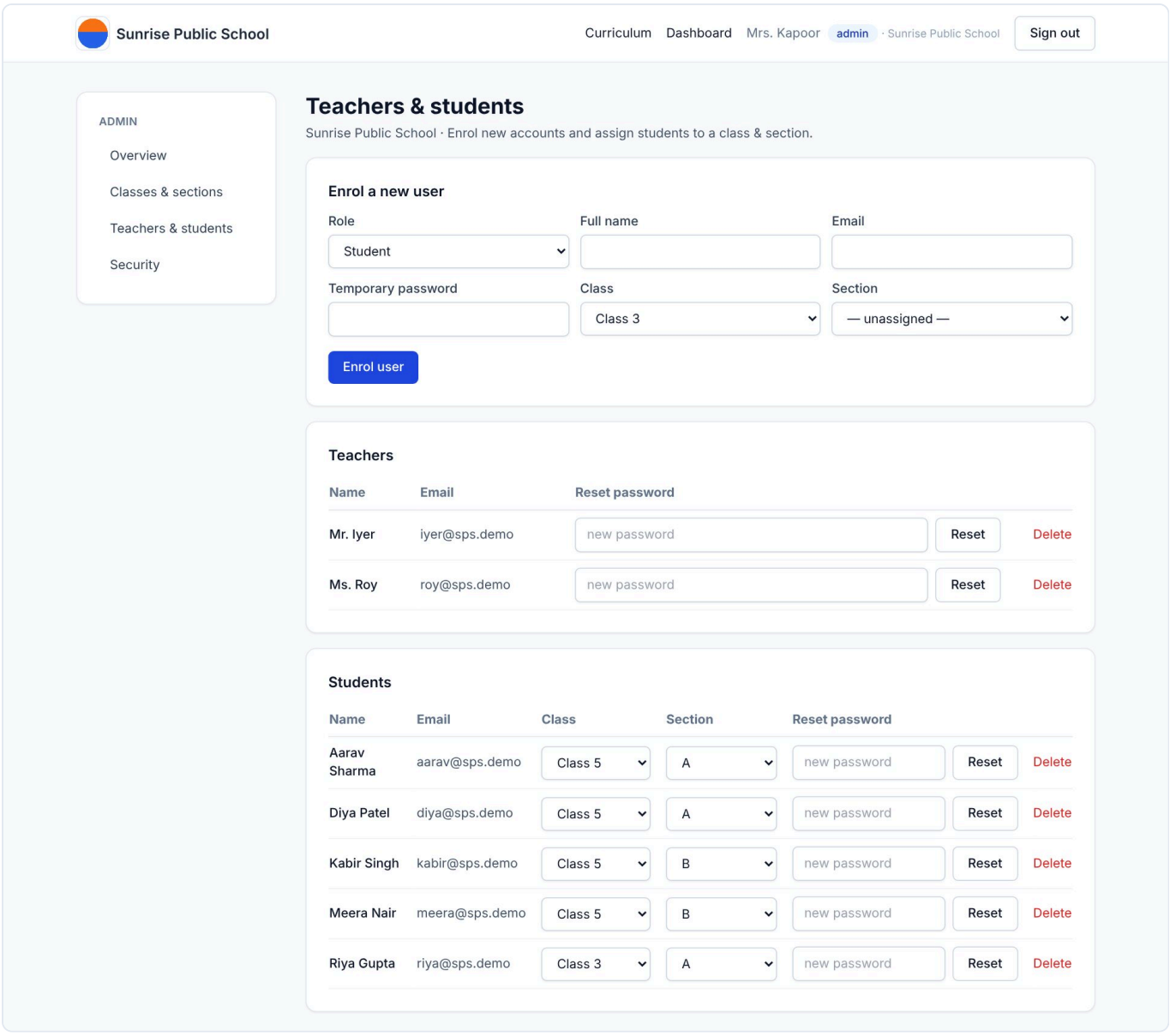
Admin overview — teachers, students and sections in this school, class by class.

Setting up a new school year — in order

1. **Create sections** under **Classes & sections** — for example Class 5 → "A", Class 5 → "B".
2. **Enrol teachers** under **Teachers & students** with a temporary password.
3. **Assign teachers to sections.** Pick a *class teacher* and tick any *other teachers* (for example a separate CT & AI teacher).
4. **Enrol students** into a class and section.
5. **Share the school link** (`/SPS-DEL`) and temporary passwords with teachers and students.



Sections — each section has one class teacher plus optional co-teachers. A teacher only ever sees the sections ticked here.



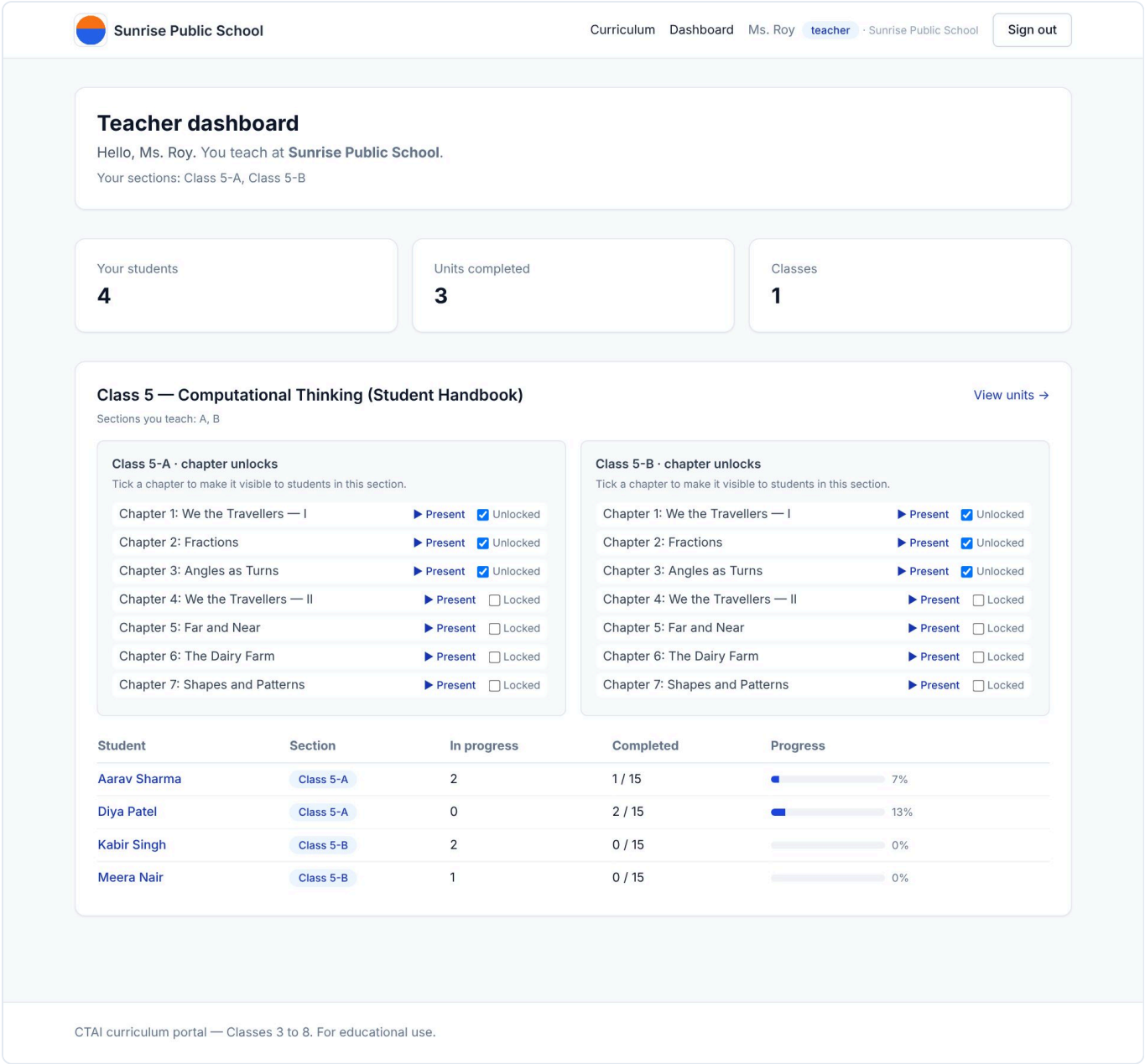
Teachers & students — enrol people, move a student to another class or section, reset a password, or remove an account.

Tip: enrolling a whole class

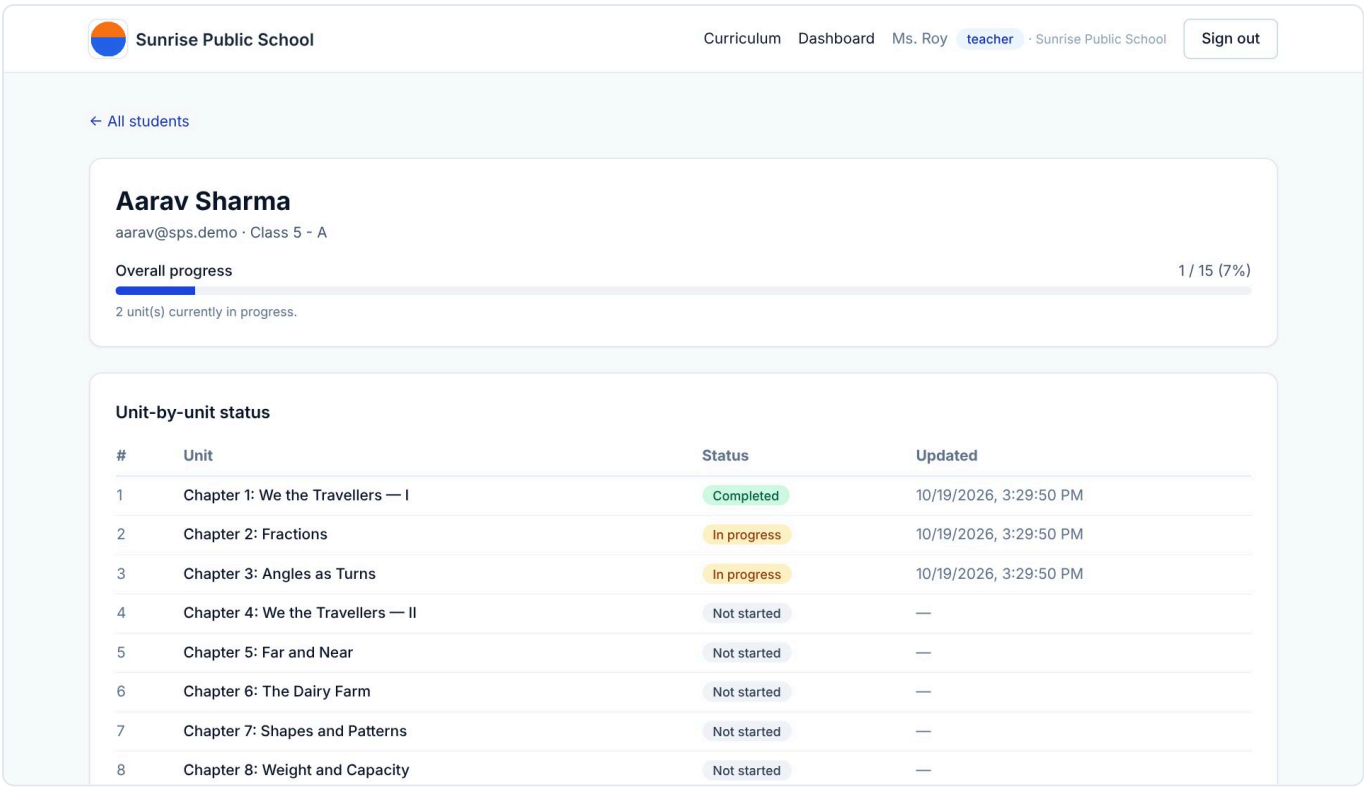
Today students are added one at a time. A CSV bulk import is on the to-do list (section 15). Until then, set up sections first so the class and section dropdowns are ready while you type.

7 Teacher

A teacher's dashboard shows only their sections. From here they open chapters to the class and follow every student's progress — marking is automatic.



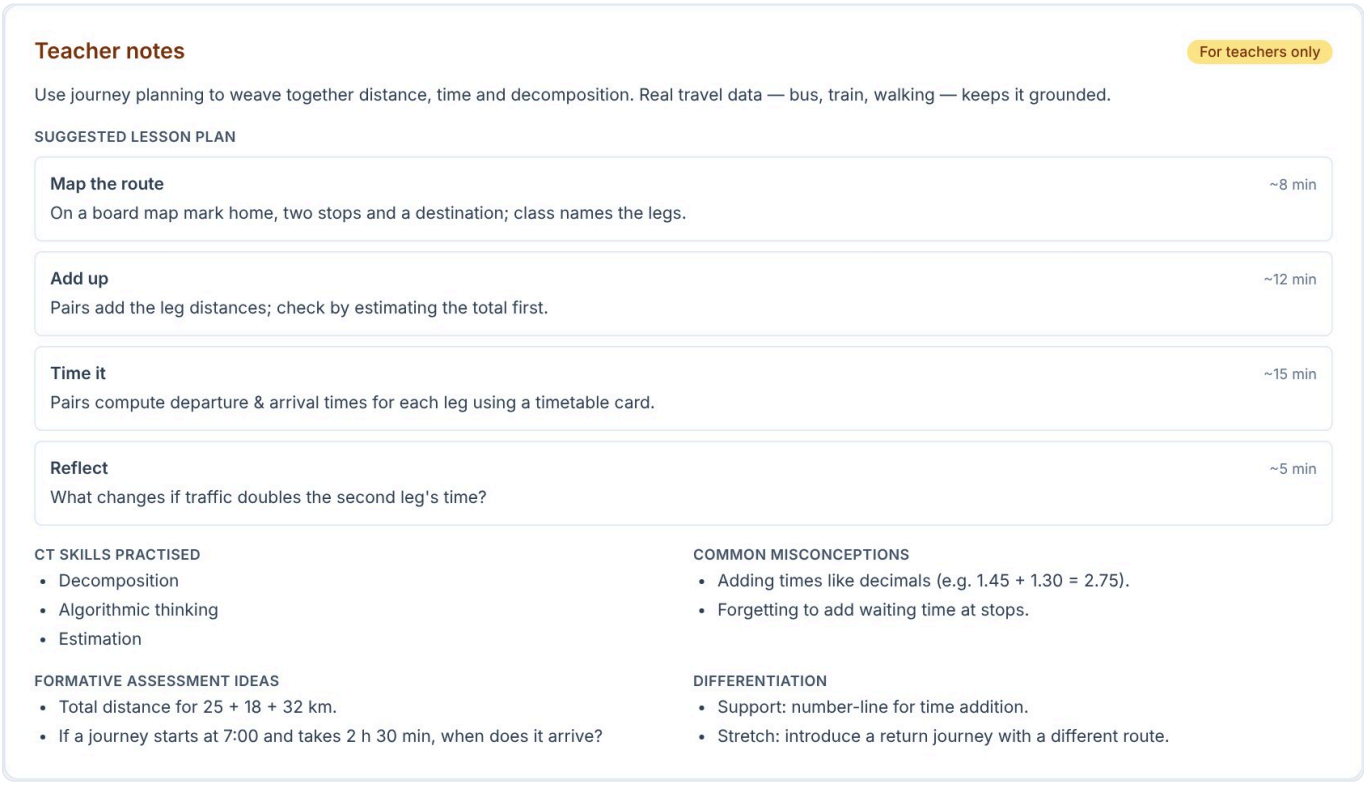
Teacher dashboard. Top: one chapter-unlock checklist per section — tick a chapter to open it for that section, or tap ▶ **Present** to show it on the class board. Bottom: each student's chapters in progress and completed, with a progress bar.



Student detail — click a student's name to see every chapter's status and when it last changed.

Teacher notes on every chapter

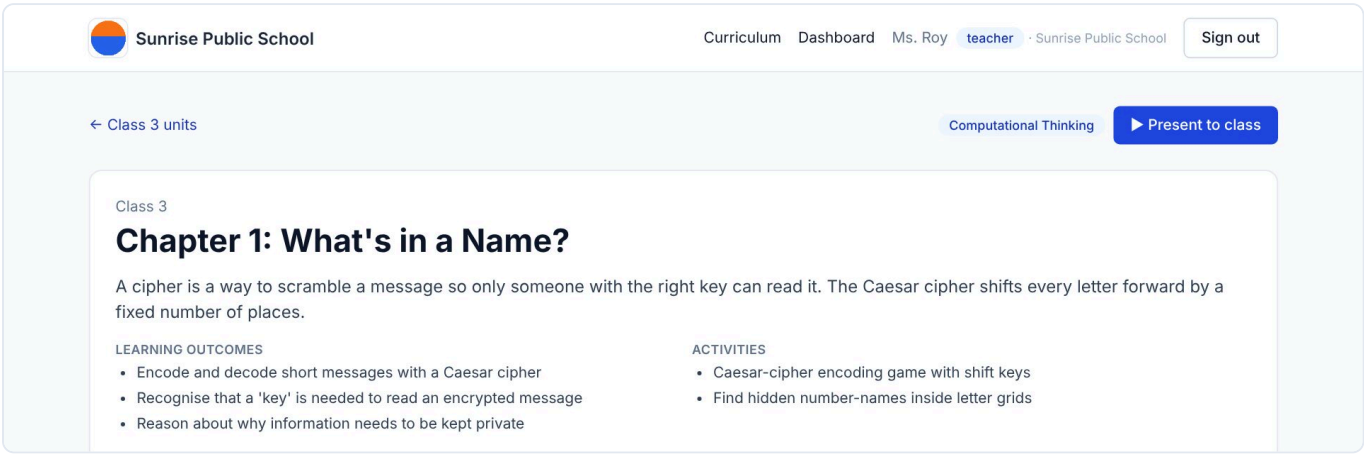
Open any chapter while signed in as a teacher and a yellow **Teacher notes** panel appears below the practice. Students never see it.



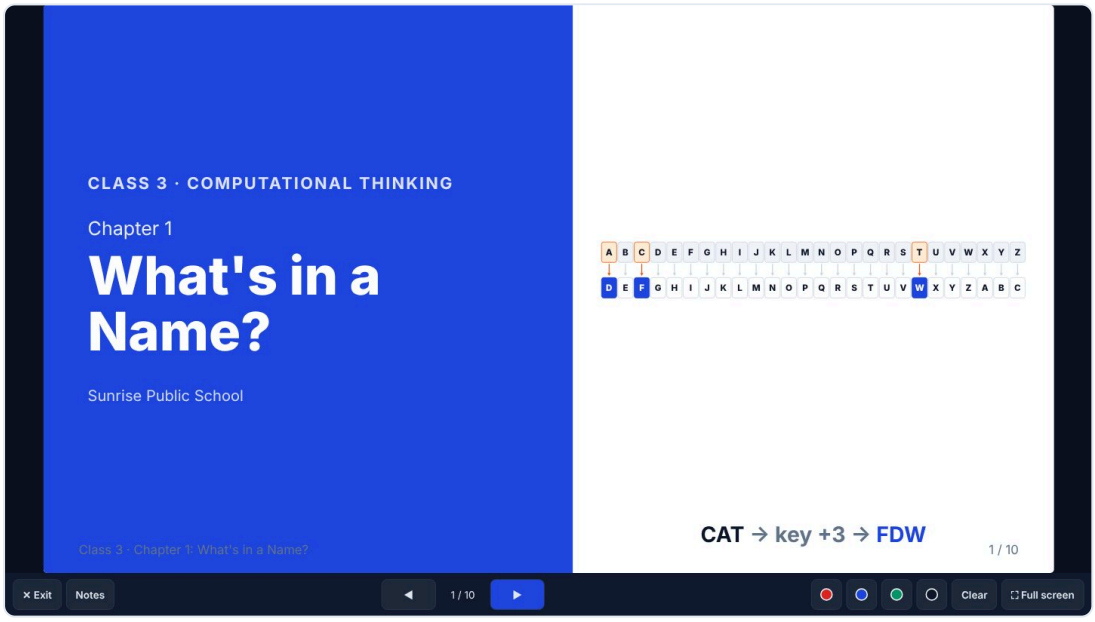
Teacher notes — a timed four-step lesson plan, the computational-thinking skills practised, common misconceptions, quick assessment ideas and support/stretch ideas.

8 Presenting on the IFP

Every one of the 80 chapters has a ready-made, full-screen slide presentation, so a teacher can explain the topic on the interactive flat panel (IFP) or a projector before students practise on their own.



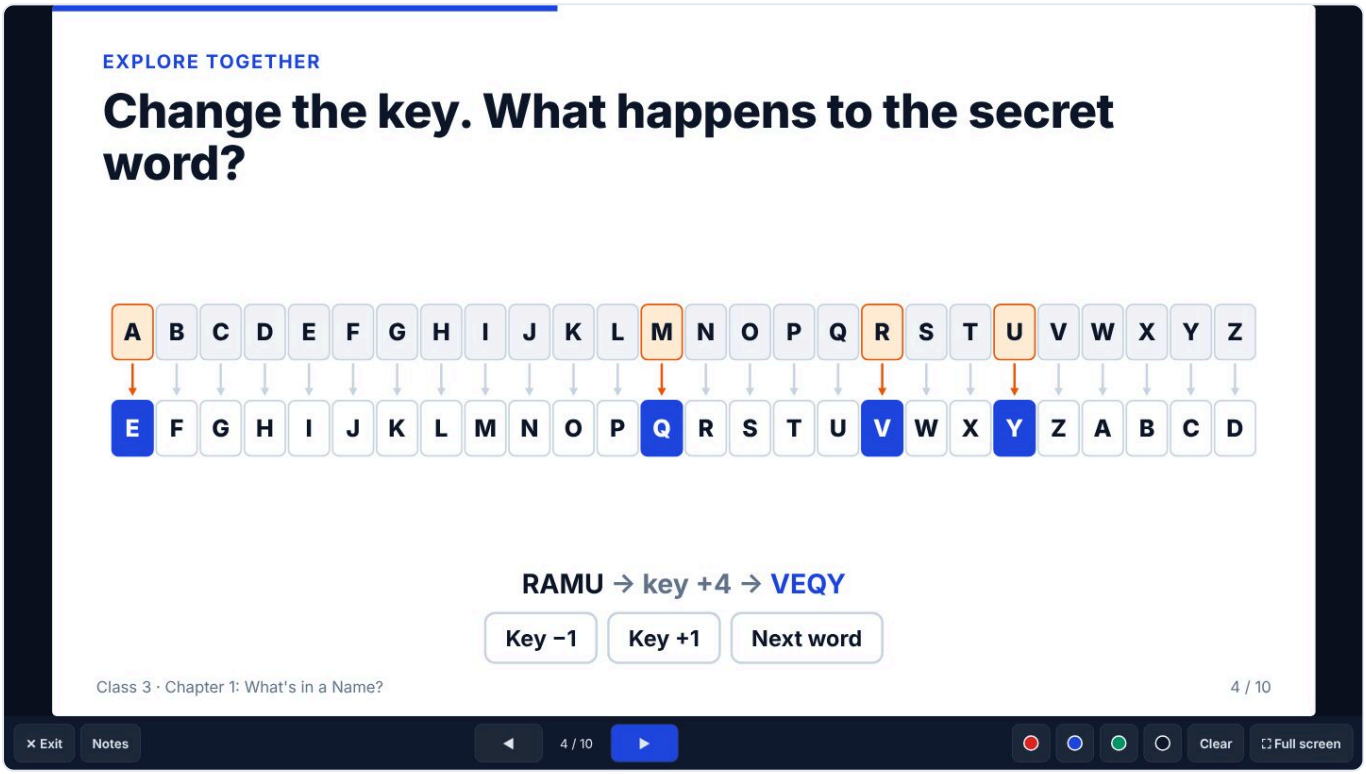
Starting a presentation. Teachers, school admins and the superadmin see ▶ **Present to class** at the top of every chapter page. The teacher dashboard's unlock lists also have a ▶ **Present** link beside each chapter. Students never see these buttons, and a student who types the address is sent back to the normal chapter page.



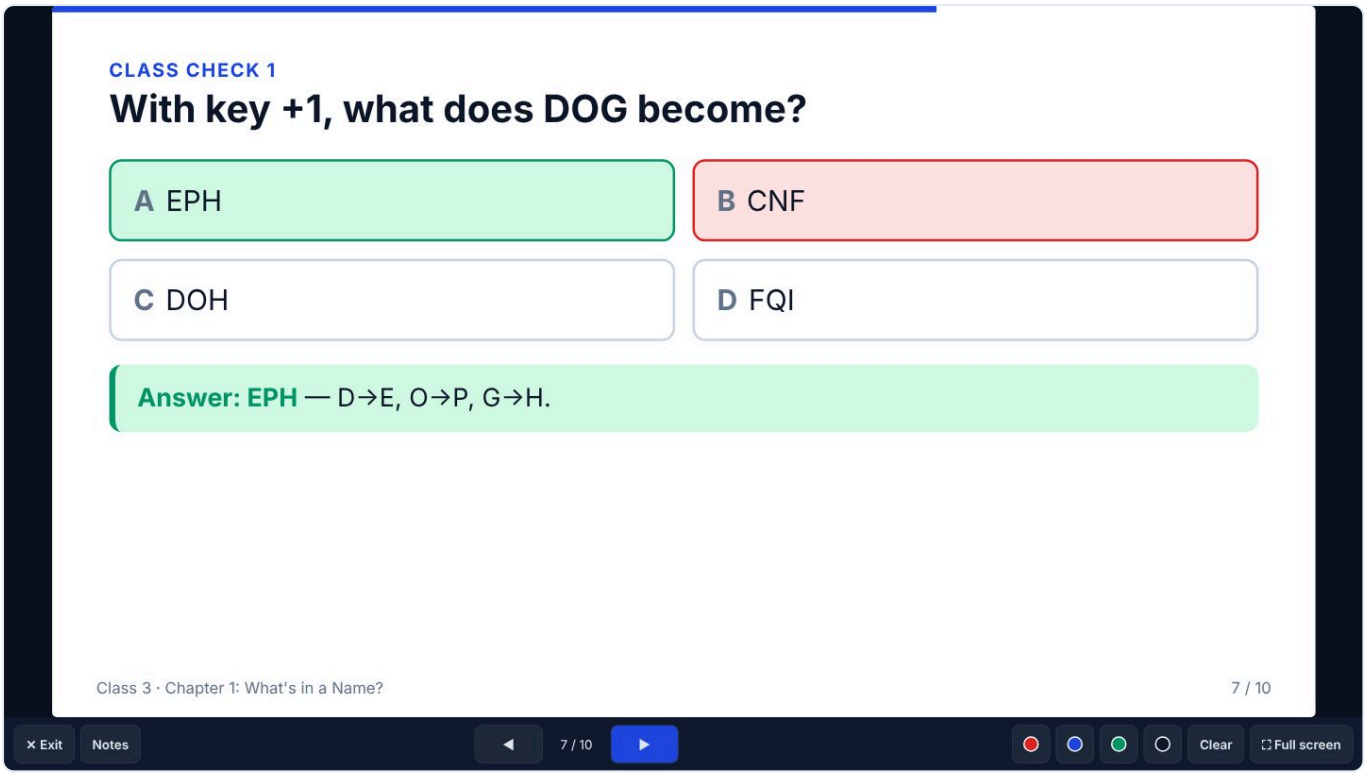
Title slide with the class, stream, chapter and school name, beside the chapter's illustration.

What is in each presentation (about 10 slides)

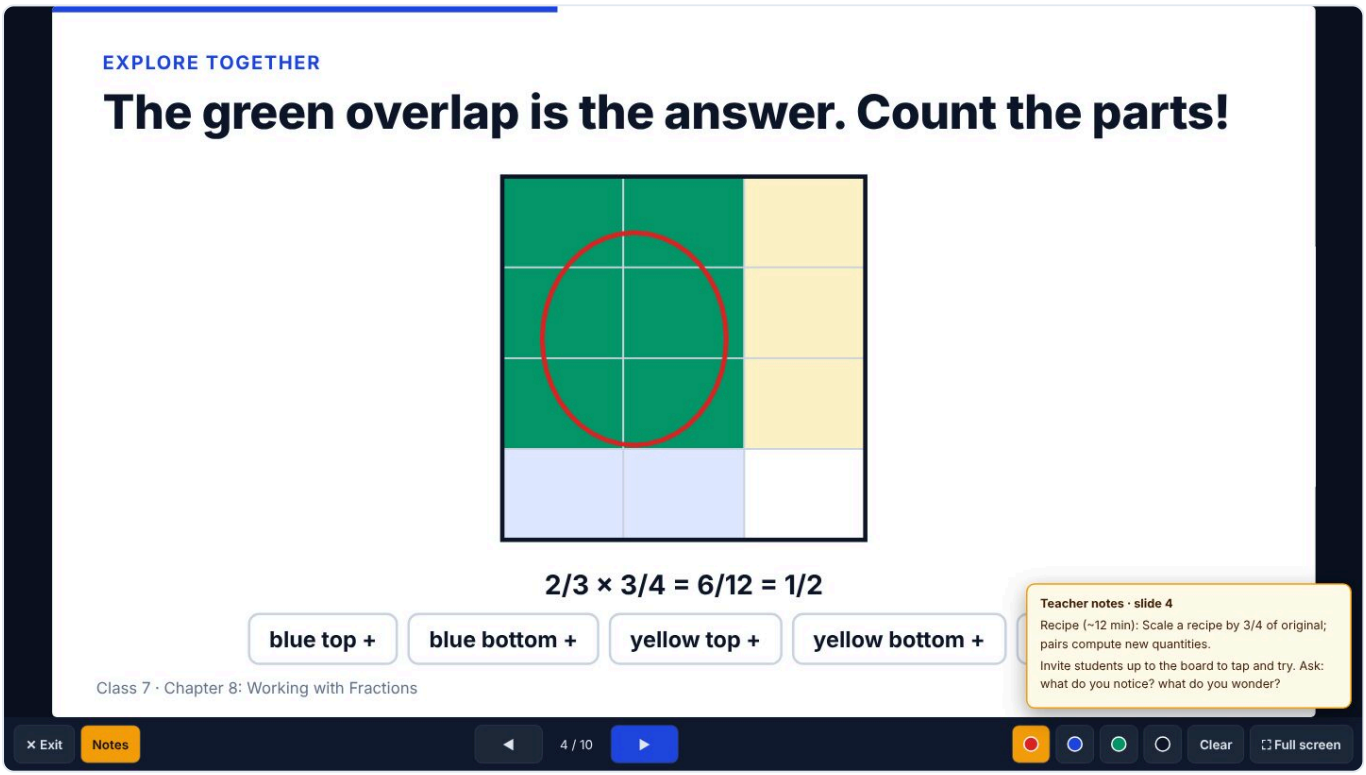
Slide	What the class sees
1. Title	Chapter name, class and school, with an illustration
2. Learning goals	"Today we will learn to…" — the chapter outcomes, numbered
3. The big idea	The concept in one sentence, plus a picture that shows it
4. Explore together	An **interactive illustration** students come up and touch — e.g. change a cipher key, build numbers with blocks, set a clock, share sweets onto plates, balance a scale, walk a map grid, tilt a transversal, light up binary bulbs
5–6. Worked examples	The chapter's examples; the solution appears only when the teacher taps *Show solution*
7–8. Class check	Two new multiple-choice questions (not the students' practice questions). Tap an option — it turns red or green on *Reveal answer*
9. Activity time	The chapter's hands-on activity
10. Recap	Key words and "✓ I can…" statements for each outcome



Explore together (Class 3, Chapter 1). Students change the key and the next word with the big buttons; the coded word updates instantly.



Class check. Students choose an option, then the teacher reveals the answer: the correct option turns green, a wrong pick turns red, and a one-line reason is shown.



Pen and teacher notes. Pick a pen colour to draw on any slide (here, circling the overlap in the fraction model). **Notes** opens a private panel with that slide's lesson-plan step, misconceptions to watch for, differentiation and exit-ticket ideas.

Controls

Do this	To
Swipe left / right, or ◀ ▶	Previous / next slide
⛶ **Full screen** or key **F**	Fill the whole board
Coloured dots	Pen on / off; **Clear** wipes this slide's drawing
Notes or key **N**	Show / hide teacher notes
Key **R**	Reveal the answer on this slide
✕ **Exit**	Back to the chapter page

Works on any board

Slides are drawn at 16:9 and scale to any screen — 65″–86″ IFPs, projectors or a laptop. Every button is large enough to tap with a finger. Nothing extra needs installing on the panel: open the portal in its browser and sign in as the teacher.

37 kinds of illustration

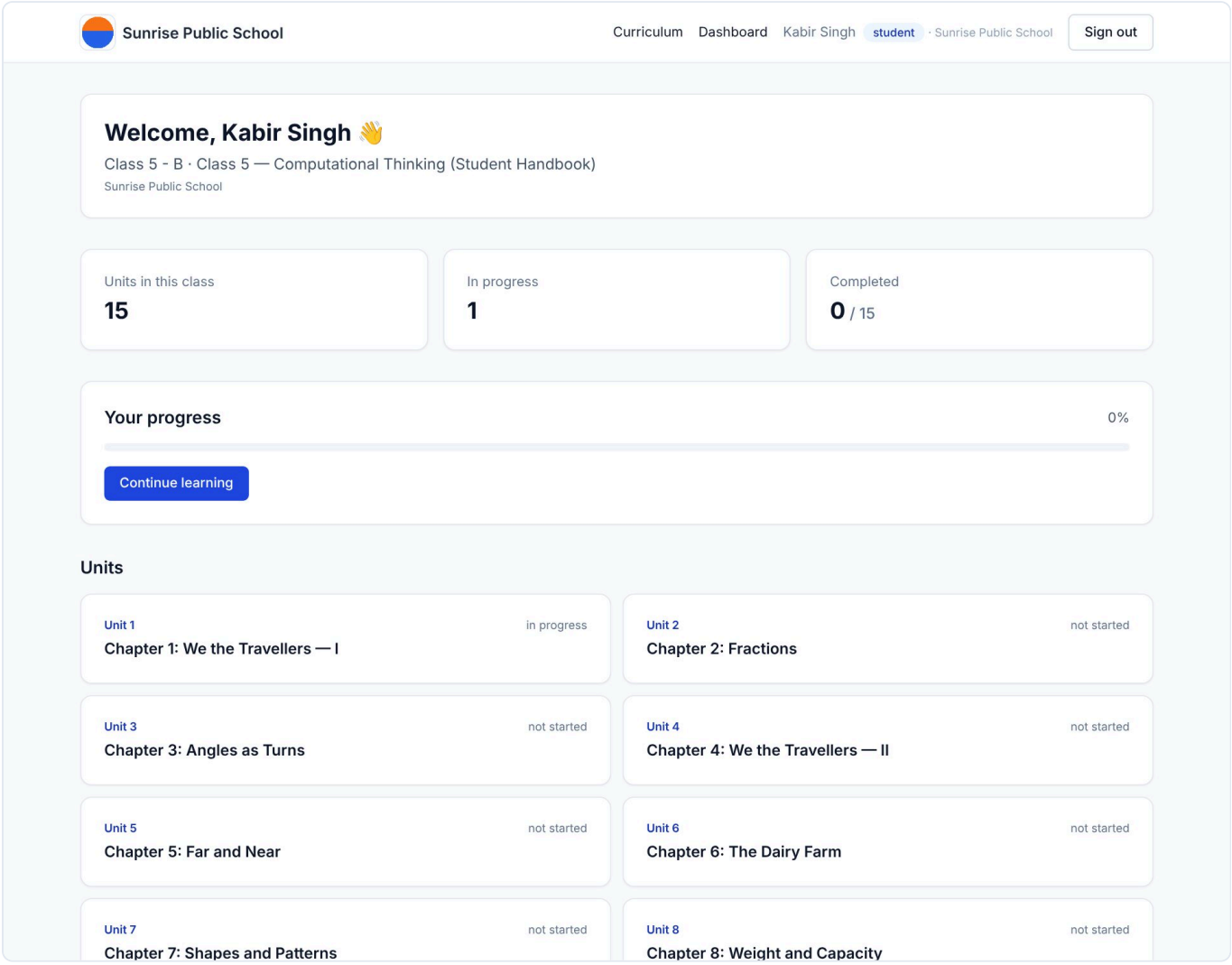
Base-ten blocks, place-value charts, number lines, fraction bars and area models, balance, money, clock, ruler and map scale, grid maps, angles, transversals, triangles, symmetry, tiling, area grids, compass construction, sieve of primes, number machines, powers, binary/Roman converters, bar/pictograph/tally/line/pie charts, clue grids, flowcharts, decision trees, AI cards, AI domains, data-bias demo and the AI project cycle. All are original drawings made for this portal.

Suggested way to use it

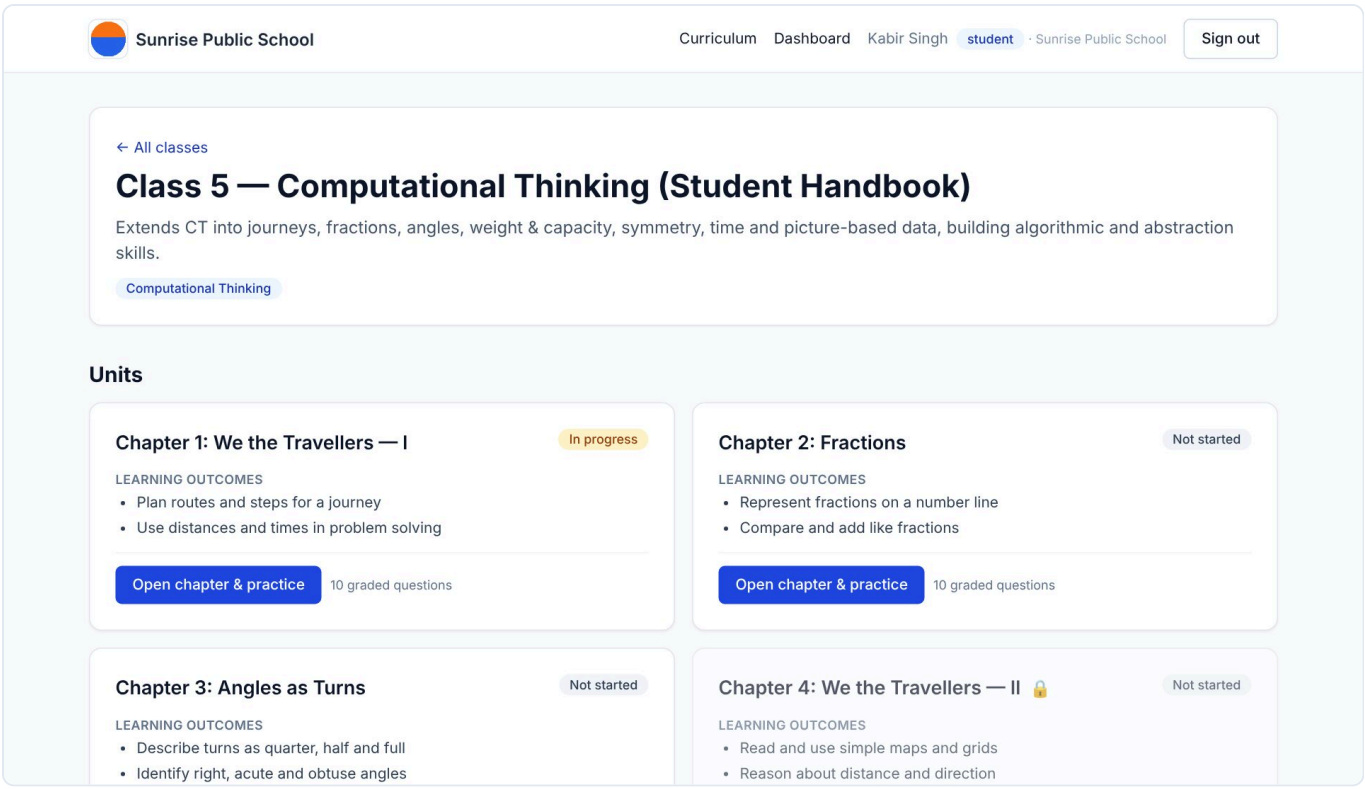
Present the chapter (about 15–20 minutes), unlock it for the section from the teacher dashboard, then let students do the self-do practice — their progress is then marked automatically.

9 Student

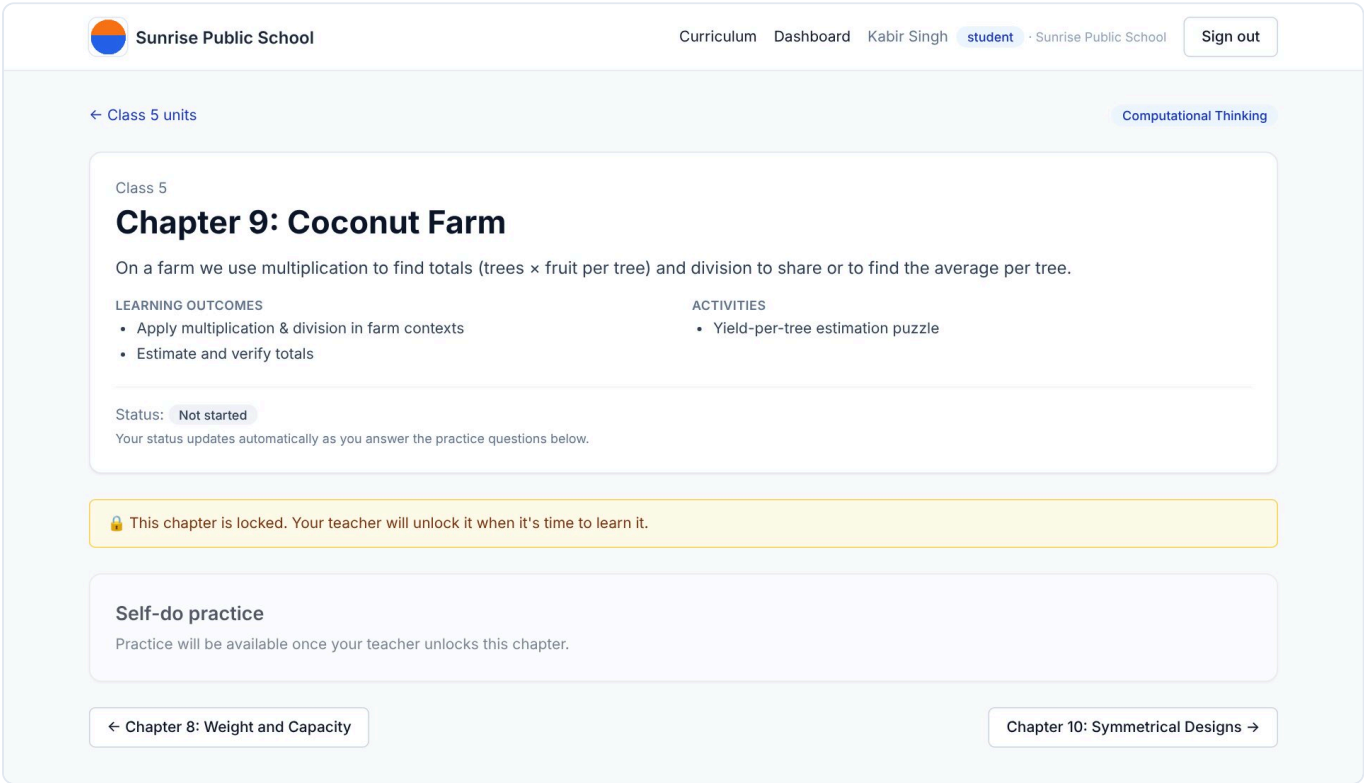
Students see only their own class. Chapters open as the teacher unlocks them, and every answer is marked the moment it is given.



Student dashboard — chapters in progress, completed, and an overall progress bar.



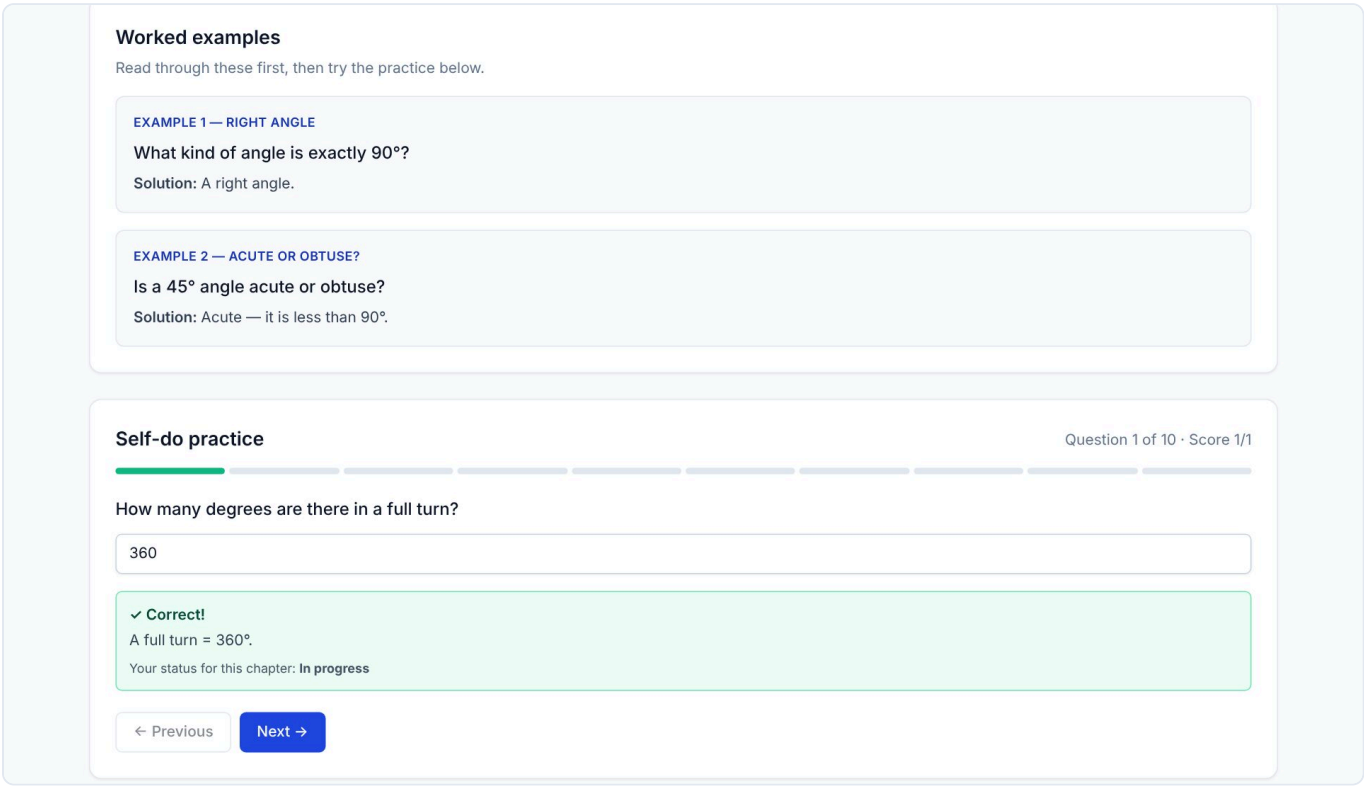
Class page. Unlocked chapters show *Open chapter & practice*; locked ones show a lock and "Waiting for teacher to unlock".



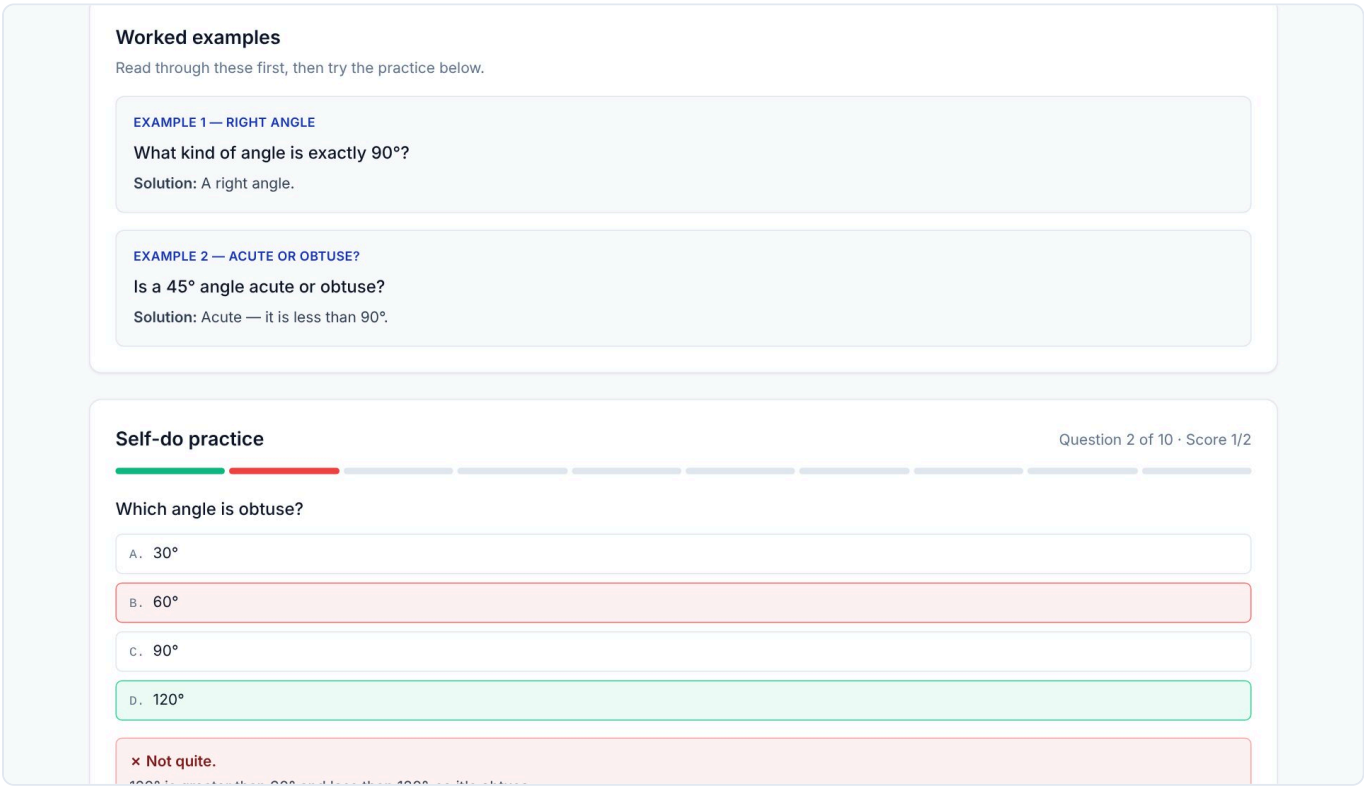
Locked chapter. Even if a student types the address directly, the examples and questions stay hidden until the teacher unlocks it.

Practising — instant right or wrong

Each chapter has **2 worked examples** and **10 practice questions** (multiple choice, true/false and short typed answers). Clicking an option marks it straight away: green for right, red for wrong, with the correct option highlighted and a one-line explanation.



A correct typed answer — the coloured bar fills question by question, and the chapter status updates underneath.



A wrong choice — the student's pick turns red, the right answer turns green, and the explanation shows why.

10 Curriculum coverage

The chapter list for every class follows the official CBSE 2026-27 CT & AI handbooks. Classes 3–5 are Computational Thinking only; Classes 6–8 add an Artificial Intelligence part.

Class	CT chapters	AI chapters	Worked examples	Practice questions	AI chapters
3	14	—	28	142	—
4	14	—	28	140	—
5	15	—	30	150	—
6	10	4	28	140	Intro to AI · Basic Data Concepts · Pattern Recognition & Decision Making · Ethics & Digital Responsibility
7	8	4	24	120	AI Domains · AI in Industries · Data Visualisation · Ethics & AI Bias
8	7	4	22	110	AI Project Lifecycle · AI Applications · Data & Fairness · Responsible AI
Total	**68**	**12**	**160**	**802**	

Every chapter contains

- **Introduction** — the idea in plain words
- **Learning outcomes** and **activities**
- **2 worked examples** with full solutions
- **10 graded questions** with explanations
- **Teacher notes**: 4-step timed lesson plan
- CT skills practised
- Common misconceptions
- Assessment and differentiation ideas
- **IFP presentation**: ~10 slides, interactive illustration, 2 class-check questions

Where the content comes from

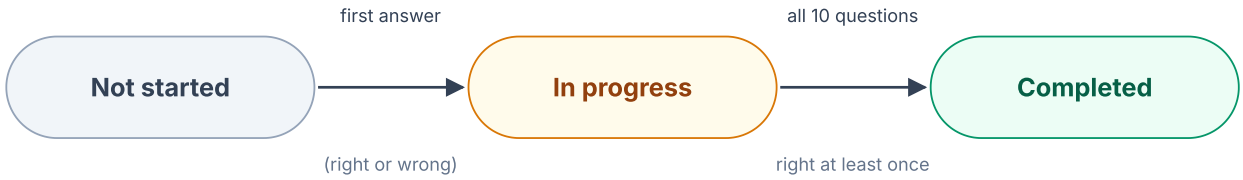
Chapter titles and topics match the CBSE handbooks. The introductions, examples, questions and teacher notes were **written originally for this portal** — they are not copied from the handbooks. Before rolling out widely, have a subject teacher review them (to-do item S5). The 12 CBSE handbook PDFs are kept in the project's `data/` folder for reference.

CBSE update check (October 2026)

The March 2026 first editions are still the current versions on cbseacademic.nic.in; no revisions or errata have been published. One chapter the earlier build had missed — **Class 6 AI Chapter 4, "Ethics and Digital Responsibility"** — has now been added.

11 How grading & progress work

Nobody marks anything by hand. Every answer is checked on the server, and the chapter's status moves forward on its own.



The rules

- **In progress** as soon as a student answers any question in the chapter.
- **Completed** when every question in the chapter has been answered correctly at least once. Wrong attempts don't undo earlier right ones, and students can retry freely.
- Progress is saved only for a **student** working in **their own class** on an **unlocked** chapter. Teachers can try the questions, but nothing is recorded.
- Every attempt (right or wrong) is stored, so detailed reports can be added later.

Typed answers are marked fairly

Short typed answers don't need to match letter-for-letter. These all count as correct when the answer is `240`, `3x+6` or `anonymisation`:

Student types	Accepted because
`240 km`	a number with a unit after it
`₹5`, `Rs 5`	currency symbol ignored
`3,600` or `1,00,000`	correct international or Indian digit grouping (but `36,00` is rejected)
`6 + 3x`, `3X+6`	spacing and capitals ignored; equivalent form listed
`4.5 × 10^3`, `4.5x10^3`	×, * and x treated alike
`anonymization`, `Human-in-the-loop`	UK/US spellings and hyphenation

Wrong answers are still wrong: `2400` for 240, `8/12` when simplest form is asked, or a different word.

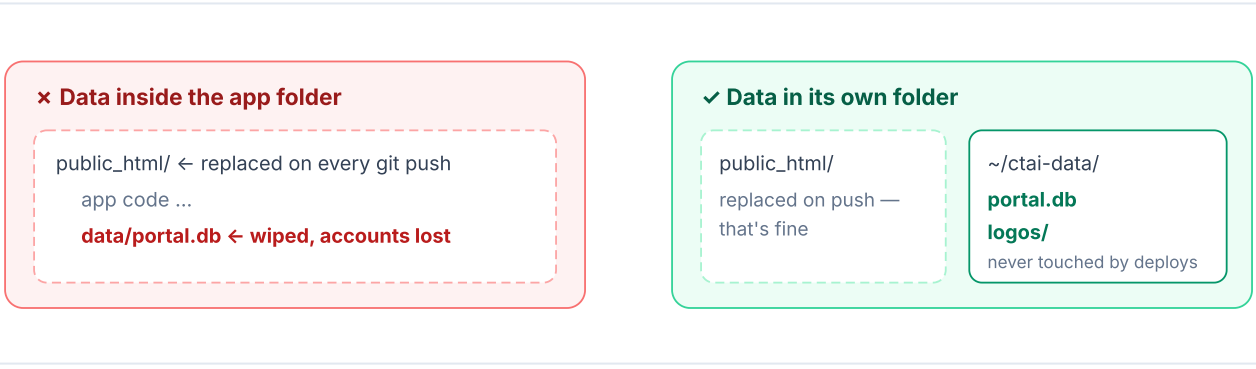
12 Security & your data

What is protected today

Protection	What it means
Hashed passwords	Passwords are scrambled with bcrypt; even someone with the database can't read them.
Secure session cookie	Sign-in is held in an encrypted, browser-only cookie that expires after 7 days.
School separation	Every admin and teacher request is checked against their school; one school can never see another's people or progress.
Hidden answer keys	Answers are checked on the server and never sent to the browser before the student answers.
Two-step sign-in	The portal owner can require a Google Authenticator code; the QR code is made on your own server.
No self-sign-up	All accounts are created by an admin; the one-time setup page locks itself after first use.

Where your data lives

Everything — schools, accounts, sections, progress, every attempt — is in a single file, `portal.db`. Uploaded logos sit next to it in `logos/`. The folder is set by `CTAI_DATA_DIR`.



Why accounts kept disappearing after each commit

Hostinger's Git deploy replaces the whole app folder on every push. The database used to live inside that folder, so each deploy erased it. Setting `CTAI_DATA_DIR` to a folder outside the app (right-hand side above) fixes this permanently — see section 13 and to-do item M1.

13 Running it on your server

Full step-by-step instructions are in `HOSTINGER.md` and `SYNOLOGY.md` in the project. The essentials:

Hostinger (Node.js app + Git deploy)

1. **Make a data folder outside the app.** Over SSH: `mkdir -p ~/ctai-data`
2. **Set environment variables** in hPanel → Node.js → Environment variables:

Name	Value
`CTAI_DATA_DIR`	full path of the folder above, e.g. `/home/u123456789/ctai-data`
`JWT_SECRET`	a long random string (e.g. from `openssl rand -hex 32`) — keep it secret, never change it casually
`NODE_ENV`	`production`

3. **Startup file** `server.js`; then **Run NPM Install**, **Run NPM Script → build**, **Restart**.
4. **Check** it's using the safe folder: `npm run db-status` must print your `~/ctai-data` path.
5. **First visit** `/register` once to create the superadmin.

After each update

Push to GitHub → Hostinger deploys → hPanel → Node.js → **Run NPM Install** (only if new packages were added — this release added `qrcode`) → **Run NPM Script → build** → **Restart**.

Synology NAS (Docker)

Put the code in `/volume1/docker/ctai-app` and the data in a separate `/volume1/docker/ctai-data`; set `CTAI_DATA_PATH` and `JWT_SECRET` in `.env`; create the project in Container Manager from the included `docker-compose.yml`.

Backups

Back up the data folder (`portal.db` + `logos/`) at least daily. The code itself is safe in GitHub.

```
# one-off backup over SSH
tar -czf ctai-backup-$(date +%F).tar.gz ~/ctai-data
```

14 What's done

Area	Feature	Status
Schools & accounts	Many schools in one installation, fully separated	Done
	Per-school name, logo and sign-in page at /<code>	Done
	Superadmin → school admin → teacher / student accounts; no self-sign-up	Done
	Sections with class teacher and co-teachers	Done
Security	Change password (superadmin, school admin); admin password resets	Done
	Google Authenticator 2-step sign-in for the superadmin	Done
	Hashed passwords, secure cookies, hidden answer keys, school isolation	Done
Teaching	Teacher sees only own sections and their students	Done
	Chapter unlock per section; locked chapters fully hidden from students	Done
	Per-student progress view	Done
	Teacher notes for all 80 chapters	Done
	“Present to class” IFP slides for all 80 chapters: interactive illustrations, tap-to-reveal examples and class checks, pen, teacher notes	Done
Learning	80 chapters, 160 worked examples, 802 questions (MCQ, true/false, typed)	Done
	Instant right/wrong on click, with explanations	Done
	Automatic progress (Not started → In progress → Completed); fair typed-answer marking	Done
Running it	Single-file SQLite database, movable via CTAI_DATA_DIR	Done
	Tools: `npm run create-admin`, `npm run db-status`	Done
	Hostinger and Synology/Docker guides	Done
Reports	Every attempt is recorded, but not yet shown as reports	Partly

Fixed while preparing this guide

Commit	Problem → fix
`42a1a75`	**Students could mark chapters "Completed" without answering** through an old hidden address → removed it.
`42a1a75`	**Owner could be locked out** if 2-step sign-in was half set up → sign-in now checks both settings.
`73dc7af`	**Practice showed only 3 of 10 questions**, so "Completed" could never be reached → all 10 now appear.
`6b3cdcb`	**The 2-step sign-in secret was sent to an outside QR website** → QR now made on your own server.
`b660162`	**Class 6 AI Chapter 4 was missing** → added with full content. **Correct typed answers were marked wrong** → fair matching.
`e0f5468`	**Locked chapters' examples visible via direct link** → now hidden until unlocked. Plus small layout fixes.

15 What still needs to be done

Listed by priority. Must = do before real students use it. Should = needed for a smooth school year. Could = nice improvements. Effort: S = hours, M = 1–3 days, L = a week or more.

#	What	Why	Effort	Who
M1	**Set `CTAI_DATA_DIR` on Hostinger** (section 13) and confirm with `npm run db-status`.	Stops accounts being wiped on every push.	S	You
M2	**Set a strong `JWT_SECRET`** on the server, and change the app to refuse to start without one.	Without it the app falls back to a built-in default secret that anyone reading the code could use to fake a sign-in.	S	You + developer
M3	**HTTPS** on the domain (Hostinger's free SSL).	Passwords must not travel unencrypted; secure cookies need HTTPS.	S	You
M4	**Automatic daily backups** of the data folder.	One file holds everything; deleting a school can't be undone.	S	You
M5	**Limit repeated wrong passwords** (lock out after several failed tries).	Stops password guessing.	S–M	Developer
S1	**Bulk import** of students and teachers from a spreadsheet (CSV).	Typing 40 students per section one by one is slow.	M	Developer
S2	**Reports**: per-question results for a section, and download as CSV/PDF.	Attempts are already stored; teachers and principals need to see weak areas.	M	Developer
S3	**Forgot-password** by email, and let teachers change their own password.	Reduces admin workload.	M	Developer
S4	**Year-end promotion**: move a whole section to the next class.	Needed every April.	M	Developer
S5	**Subject-teacher review** of all 802 questions, examples and teacher notes.	Content is original and checked for format errors, but not yet reviewed by an educator.	L	Teachers
S6	**Lock markers on the student dashboard** and links straight to each chapter.	The dashboard list doesn't yet show which chapters are locked.	S	Developer
S7	**2-step sign-in for school admins**.	Admins can reset every password in their school.	S	Developer
S8	**Automated tests** run on every push.	Most bugs fixed this round would have been caught automatically.	M	Developer

#	What	Why	Effort	Who
C1	**Pictures and diagrams** in questions and examples.	The CBSE handbooks are very visual (shapes, maps, pictographs). The IFP slides already have illustrations; the students' practice questions do not yet.	L	Developer + teachers
C2	**Hindi / bilingual** content and screens.	Wider reach across schools.	L	Translator + developer
C3	**Assignments and timed tests** with due dates.	Formal assessment, not just practice.	M–L	Developer
C4	**Badges and streaks** for students.	Motivation for younger classes.	M	Developer
C5	**Install as a phone app** (PWA) and offline reading.	Easier access on shared or low-connectivity devices.	M	Developer
C6	**Activity log** of admin actions (who deleted or reset what).	Accountability across schools.	M	Developer
C7	**Postgres option** for hosts without a permanent disk (e.g. Vercel).	Only needed if you leave Hostinger/Synology.	M	Developer

Suggested order

This week: **M1–M4** (all quick server settings). Next: **M5, S6, S7** (small code changes), then **S1 and S2** before term starts. **S5** can run in parallel with teachers.

16 Quick reference

Addresses

Address	Who	What
/	Everyone	Home page, class list
/curriculum	Everyone	Browse classes and chapter outlines
/<SCHOOL-CODE>	School users	Branded sign-in, e.g. /SPS-DEL
/login	Superadmin	Owner sign-in (with 2-step code if enabled)
/register	First visit only	Creates the superadmin once, then switches off
/superadmin	Superadmin	Schools, school admins, security
/admin	School admin	Sections, teachers & students, security
/dashboard/teacher	Teacher	Unlocks and student progress
/curriculum/<class>/<chapter>/present	Teacher, admin	Full-screen class presentation, e.g. /curriculum/5/c5-ct-3/present
/dashboard/student	Student	Own progress and chapters

Server commands

```
# show where the database is and what it holds
npm run db-status

# create or reset the superadmin from the command line
npm run create-admin -- --role superadmin --email you@example.com --password 'YourStrongPass1'

# create or reset a school admin
npm run create-admin -- --role admin --school SPS-DEL --email principal@example.com --password
'AnotherPass1'
```

Password rules

Account	Minimum length	Who can reset it
Superadmin	8	Self (Security page) or the create-admin command
School admin	8	Self, or the superadmin
Teacher, student	6	Their school admin

17 Glossary

CT	Computational Thinking — breaking problems down, spotting patterns, step-by-step methods.
AI	Artificial Intelligence — taught in Classes 6–8.
School code	Short name for a school, e.g. SPS-DEL; also its sign-in address.
Section	A class group such as 5-A, with its own teachers and chapter unlocks.
Unlock	A teacher opening a chapter for one section.
IFP	Interactive flat panel — the large touch screen in a classroom. “Present to class” mode is designed for it.
Auto-progress	Chapter status updating itself from the student's answers.
2FA / two-step sign-in	A 6-digit code from Google Authenticator needed in addition to the password.
portal.db	The single database file holding all data.
CTAI_DATA_DIR	Server setting telling the app where portal.db and logos live.
JWT_SECRET	Server setting used to sign sign-in cookies; must be long, random and private.